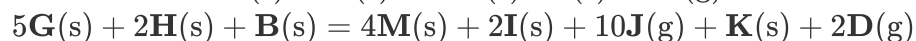
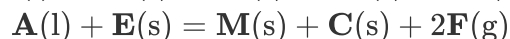
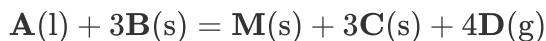


Question

People have discovered that a certain substance **M** can be prepared through the following reactions:



M is composed of two short-period elements **X** and **Y** (the atomic number of **X** is less than that of **Y**) and does not contain hydrogen. In **A**, $\omega(\text{Br}) = 95.7\%$, **B** is a sodium salt that is relatively stable at room temperature but decomposes into a large amount of gas upon heating. **E** is slightly more stable than **B**, where $\omega(\text{Na}) = 58.9\%$. **G** decomposes upon heating to release an acidic gas and a basic gas, and **G** contains a coordinate covalent bond. **I** is a Mg-containing salt, $\omega(\text{Mg}) = 25.5\%$.

The following statements about the 12 substances **A — K, M** are correct:

A. All other options are incorrect

B. 12 kinds of substances contain a total of 8 kinds of elements.

C. 12 kinds of substances contain a total of 9 kinds of elements.

D. Among 12 substances, there are 9 binary compounds.

E. Of 12 substances, there are 3 elements.

F. Of the 12 substances, 4 are gases at room temperature and pressure.

G. Of the 12 substances, three have a molecular weight exceeding 100.

H. Of the 12 substances, 3 have a molecular weight below 30.

Answer

Correct Answer: **H**

Detailed Explanation

The main body starts from **A** with $\omega(\text{Br}) = 95.7\%$:

Since the mass fraction is large, **A** is likely to contain multiple Br. Assume it contains n :

If n is 1, the remaining molecular weight $= 79.9/0.957 - 79.9 = 3.6$, no reasonable element.

If n is 2, the remaining molecular weight $= (2 \times 79.9)/0.957 - 2 \times 79.9 = 7.2$, no reasonable element.

If n is 3, the remaining molecular weight $= (3 \times 79.9)/0.957 - 3 \times 79.9 = 10.8$, which is element B.

If n is 4, the remaining molecular weight $= (3 \times 79.9)/0.957 - 4 \times 79.9 = 14.3$, no reasonable element.

If n is 5, the remaining molecular weight $= (5 \times 79.9)/0.957 - 5 \times 79.9 = 18$, no reasonable element. A larger n is unlikely to exist.

Therefore, **A** is BBr_3 ;

CHECKPOINT

2 PTS

Infer **A** is BBr_3 based on mass fraction

Then start from **E**; $\omega(\text{Na}) = 58.9\%$, considering that most sodium salts are of the 1:1 or 1:2 type, assume **E** contains one Na, the remaining molecular weight is $23.01/0.589 - 23.01 = 16.06$, which is exactly the molecular weight of NH_2^- ; therefore **E** is NaNH_2 .

CHECKPOINT

1 PTS

Infer **E** is NaNH_2 based on mass fraction

B is less stable than **E** and releases gas upon heating, and is likely to contain nitrogen; therefore **B** is sodium azide NaN_3 , which decomposes into nitrogen gas upon heating.

CHECKPOINT

1 PTS

B is sodium azide NaN_3

NaNH_2 and BBr_3 in a 1:1 ratio can synthesize **M**, BBr_3 and NaN_3 in a 1:3 ratio can also synthesize **M**; and **M** does not contain hydrogen; it is sufficient to determine that elements **X**, **Y** in **M** are B, N respectively, and according to the equation ratio, **M** is BN.

CHECKPOINT

1 PTS

Determine that elements **X**, **Y** in **M** are B, N respectively

CHECKPOINT

2 PTS

M is BN according to the equation ratio

From this, the chemical equations for the first two synthesis reactions can be written:



CHECKPOINT

1 PTS



CHECKPOINT

1 PTS



Therefore **C** is NaBr, **D** is N₂, **F** is HBr.

CHECKPOINT

0.5 PTS

C is NaBr

CHECKPOINT

0.5 PTS

D is N₂

CHECKPOINT

0.5 PTS

F is HBr

G can be used to synthesize BN, so it must contain nitrogen or boron; **G** decomposes to produce both alkaline and acidic gases and has a coordinate bond, and the only alkaline gas containing nitrogen or boron is ammonia; therefore **G** contains (NH_4^+) , so **G** can only be NH_4Cl or NH_4Br or NH_4I .

CHECKPOINT

1 PTS

G contains (NH_4^+)

According to **I** being a Mg containing salt, $\omega(\text{Mg}) = 25.5\%$, assume **I** contains only one Mg, the remaining molecular weight is 70.9, which is exactly two chlorine elements; therefore **I** is MgCl_2 , and at this time it can be determined that **G** can only be NH_4Cl .

CHECKPOINT

1 PTS

Infer **I** is MgCl_2 based on mass fraction

CHECKPOINT

1 PTS

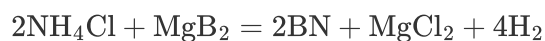
G can only be NH_4Cl

G can only be NH_4Cl , then from the third equation, **H** must contain Mg, B, according to the 2:1 reaction ratio of **G** and **H**, **H** is MgB_2 , write the equation:

CHECKPOINT

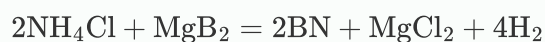
1 PTS

H is MgB_2 according to the 2:1 reaction ratio of **G** and **H**



CHECKPOINT

1 PTS



Therefore **J** is H_2 .

CHECKPOINT

1 PTS

J is H_2

Most of the reactants and products of the fourth equation are already known, try to write the equation:



According to the law of conservation of elements, **K** is NaCl .

CHECKPOINT

1 PTS

K is NaCl according to the law of conservation of elements.

The equation is:



CHECKPOINT

1 PTS



Summary: Unknown species **A – K, M** are
 BBr₃, NaN₃, NaBr, N₂, NaNH₂, HBr, NH₄Cl, MgB₂, MgCl₂, H₂, NaCl, BN respectively.

CHECKPOINT

1 PTS

A – K, M are BBr₃, NaN₃, NaBr, N₂, NaNH₂, HBr, NH₄Cl, MgB₂, MgCl₂, H₂, NaCl, BN

According to the chemical formulas, only option H is correct.